

Deep Learning

Assignment-05 (Report)

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1 Github Repository URL

https://github.com/mubashar-ds/DL_Assignment5_SimCLR_MSDS24031

2 Conceptual Warm-Up

1. In supervised classification, what information does the model learn from?

Model learns from the input images and their corresponding labels, and the labels provide correct target class and guide parameter updates during the training

2. If CIFAR-10 labels are removed, do the images still contain useful visual structure?

Yes, the images still contain meaningful information such as textures, colors, object structures, and shapes, these patterns can be learned without the labels

3. If two augmented versions are created from the same original image, should their feature representations be similar or different?

They should be similar because both views represent same underlying object

4. If two images come from different original images, should their feature representations be similar or different?

They should generally be different because they represent different visual content

5. Why might a model trained without labels still be useful for classification later?

Self supervised model can learn useful feature representations from the unlabeled data which can later be used for classification tasks with the limited labeled data

6. What is the difference between pretraining and fine-tuning?

The pretraining learns general purpose representations from the unlabeled data, while fine tuning adapts those representations to specific labeled task

3 Experimental Setup

Item	Value Used
Batch Size	64
SimCLR Epochs	50
Linear Probe Epochs	20
Fine-Tuning Epochs	20
Learning Rate	0.0003
Optimizer	Adam
Temperature	0.5
GPU/CPU Used	Colab T4 GPU
Approximate Training Time	2 hours 15 minutes approx

Table 1: Experimental Configuration

4 Supervised Baseline

ResNet18 classifier was trained using only 10% labeled CIFAR-10 data

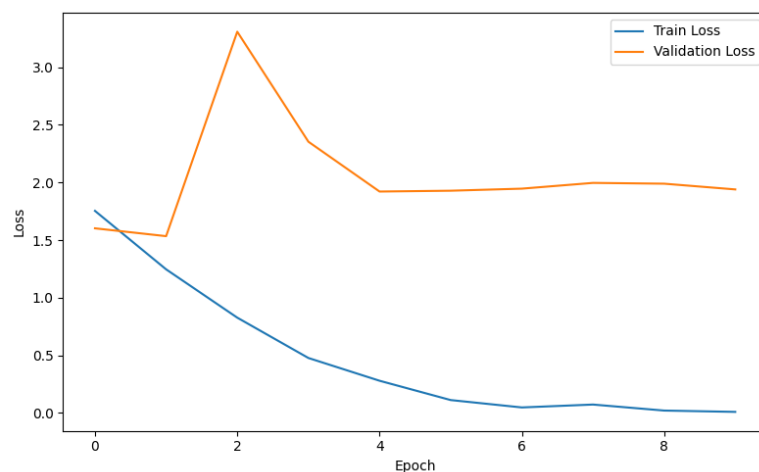


Figure 1: Training and Validation Loss for Supervised Baseline

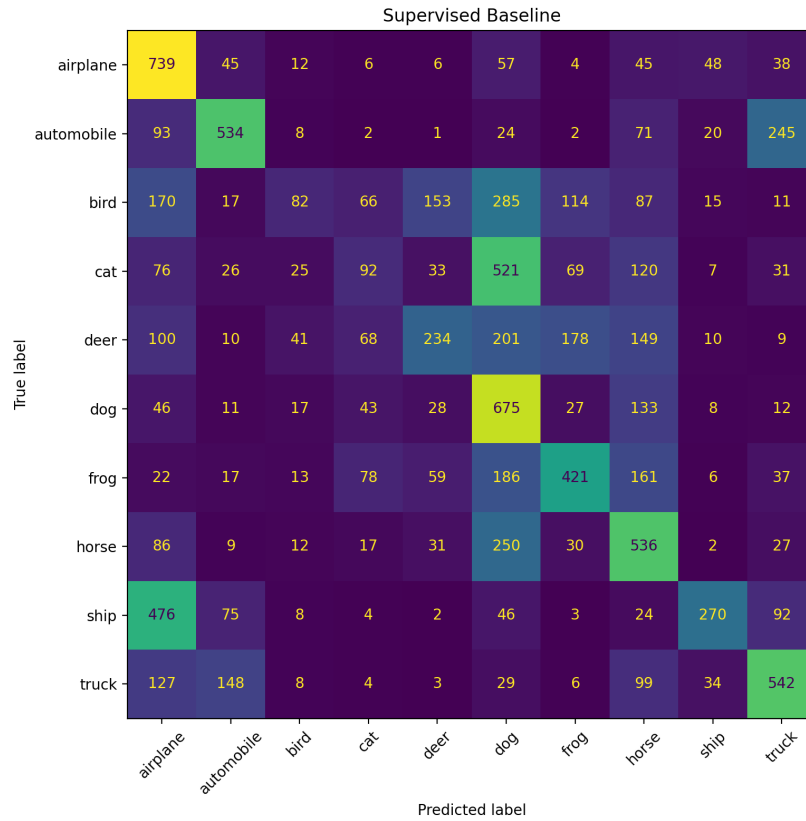


Figure 2: Confusion Matrix for the Supervised Baseline

Results

Metric	Value
Test Accuracy	45.87%

Table 2: Supervised Baseline Performance

Observations

Training loss decreased rapidly, while validation loss remained relatively high and unstable, so this shows overfitting due to limited amount of labeled data

5 SimCLR Data Augmentation

Two augmented views of the same image were generated using random cropping, color jittering, grayscale conversion, and horizontal flipping.

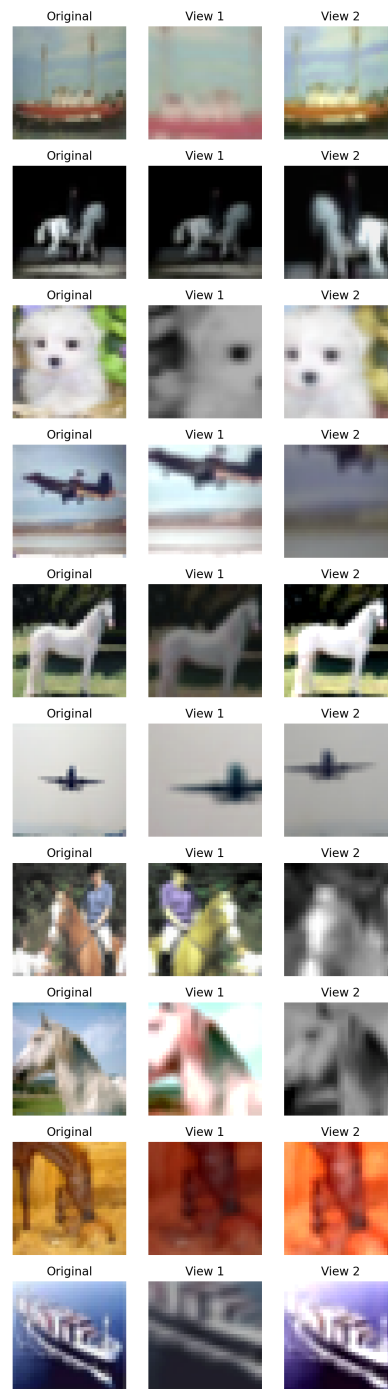


Figure 3: Examples of SimCLR Augmentations

Are the two augmented views identical?

No, random augmentations create noticeable differences between two views

Do they still represent the same object?

Yes, despite visual changes, both views contain same semantic content

Why should SimCLR treat them as a positive pair?

Because both views originate from same image and should produce similar representations

What could go wrong if augmentations are too weak?

Task becomes too easy and model may learn trivial low level features

What could go wrong if augmentations are too strong?

Important object information may be removed, thus making the positive pairs difficult to recognize.

6 Similarity Analysis Before Training

Feature similarities were measured using randomly initialized encoder

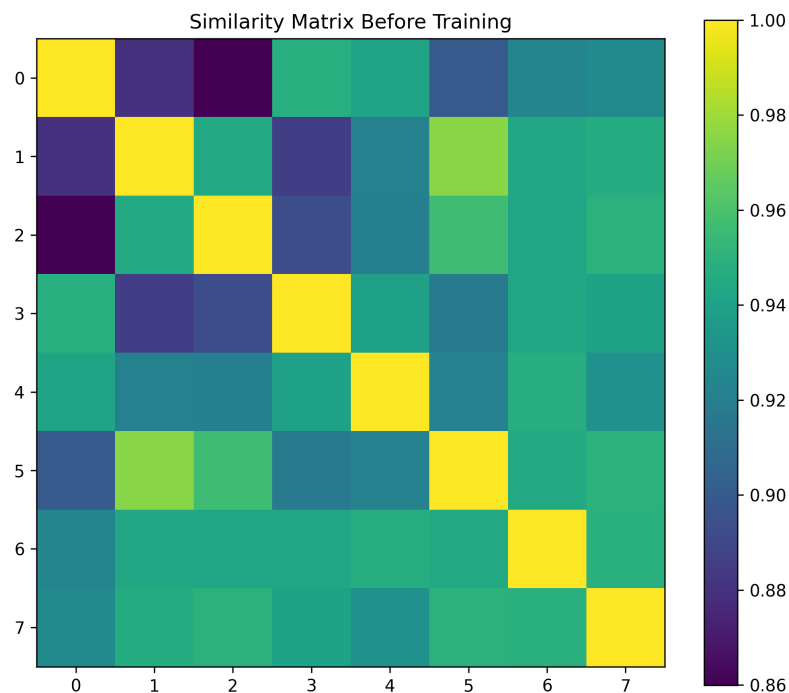


Figure 4: Similarity Matrix Before Training

Results

Comparison	Similarity
Same Image Views	0.992
Different Images	0.988

Table 3: Feature Similarities Before Training

Why is the diagonal ignored?

Diagonal compares each sample with itself, and so always has maximum similarity

Where are the positive pairs located?

Positive pairs correspond to the two augmented views of same image

Why are all other entries treated as negatives?

They correspond to the different images, and should be pushed apart during the contrastive learning

7 SimCLR Pretraining

Encoder was pretrained using NT-Xent contrastive loss for 50 epochs

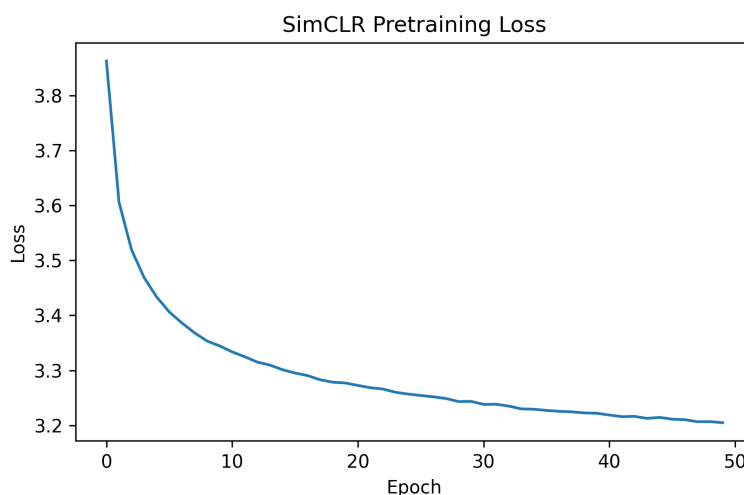


Figure 5: SimCLR Pretraining Loss

Observation

Loss decreased steadily throughout training showing successful optimization of contrastive objective

8 Similarity Analysis After Training

Feature similarities were measured again after SimCLR pretraining

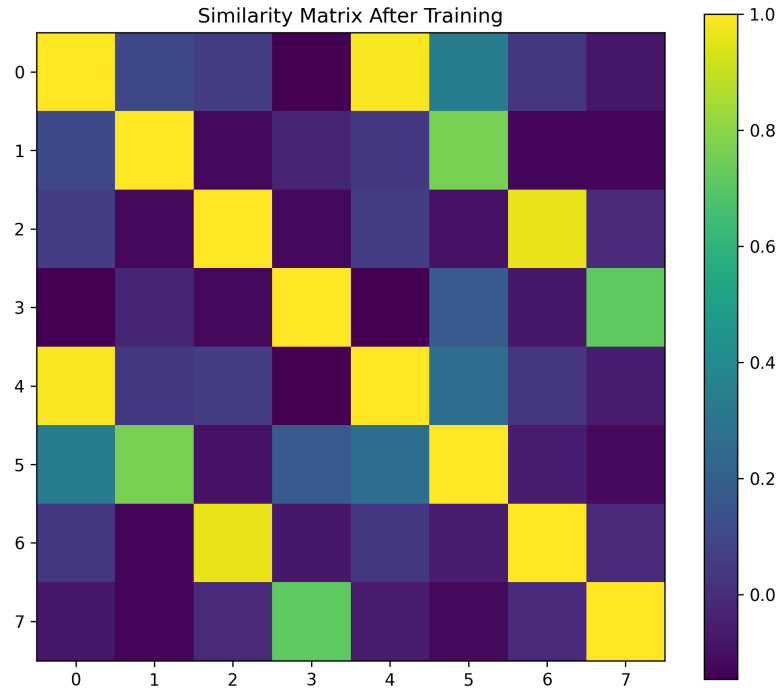


Figure 6: Similarity Matrix After Training

Results

Comparison	Similarity
Same Image Views	0.905
Different Images	0.462

Table 4: Feature Similarities After Training

Analysis

After training, similarity between the different images dropped substantially, while augmented views of same image remained highly similar, so it shows SimCLR successfully learned the discriminative feature representations

9 Linear Probe Evaluation

Linear classifier was trained while keeping encoder frozen

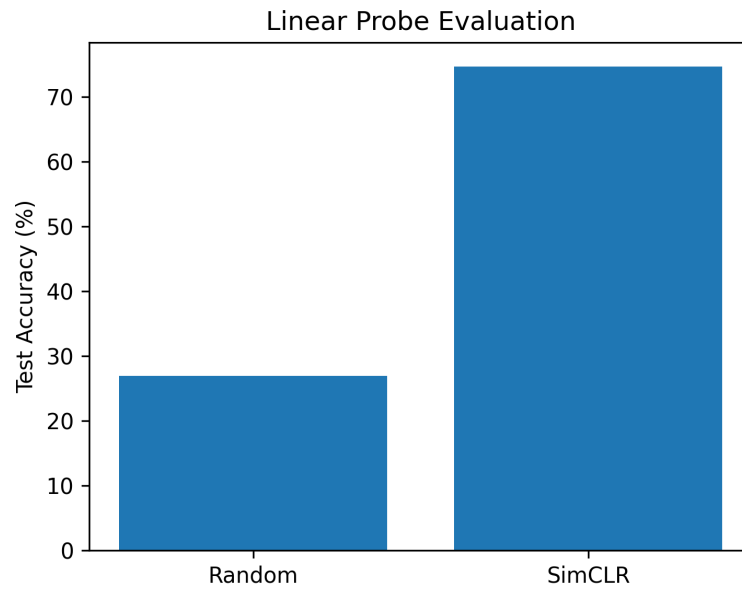


Figure 7: Linear Probe Evaluation

Results

Encoder	Test Accuracy
Random Encoder	26.960%
SimCLR Encoder	74.660%

Table 5: Linear Probe Results

Observation

SimCLR encoder significantly outperformed random encoder, so showing that useful representations were learned during self supervised pretraining

10 Fine Tuning

Pretrained encoder, and classifier were jointly optimized using labeled data

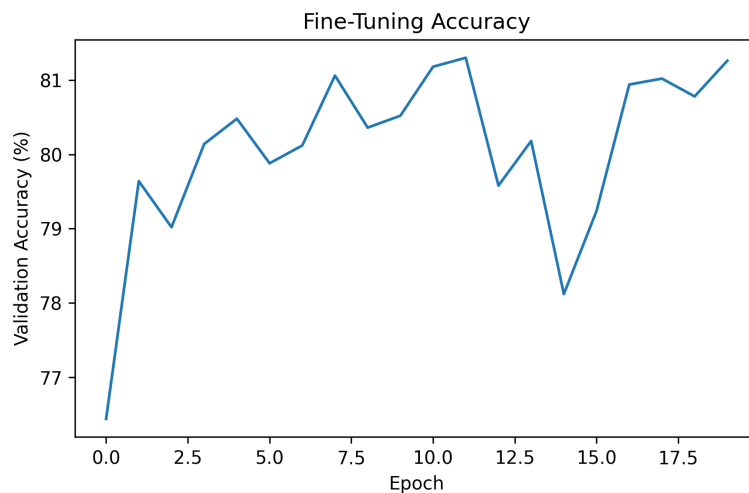


Figure 8: Fine-Tuning Validation Accuracy

Results

Model	Test Accuracy
SimCLR Linear Probe	74.660%
SimCLR Fine-Tuning	81.150%

Table 6: Fine-Tuning Results

Observation

Finetuning improved the performance by approximately 6.5 percentage points, thus showing that pretrained features provide strong initialization for supervised learning

11 PCA Feature Visualization

PCA was applied to the 512-dimensional features extracted from 1000 validation images

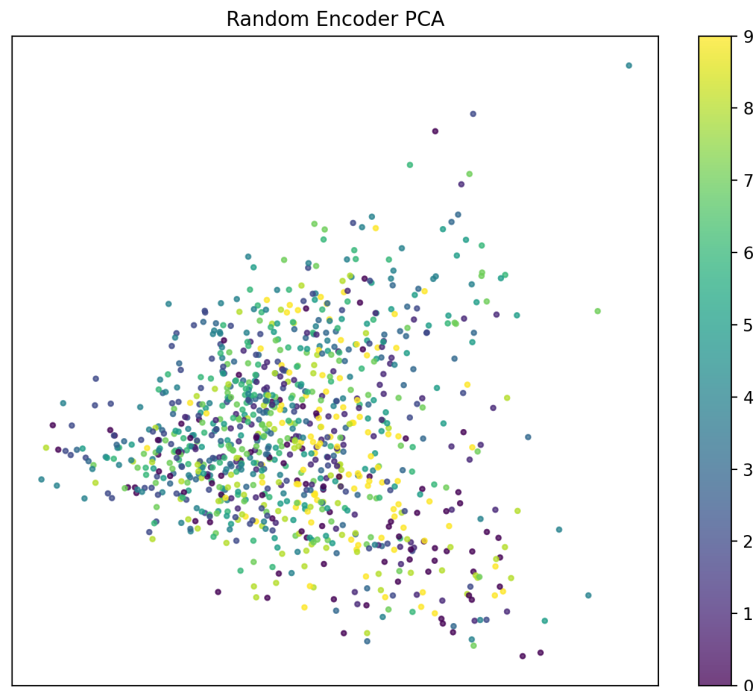


Figure 9: Random Encoder PCA

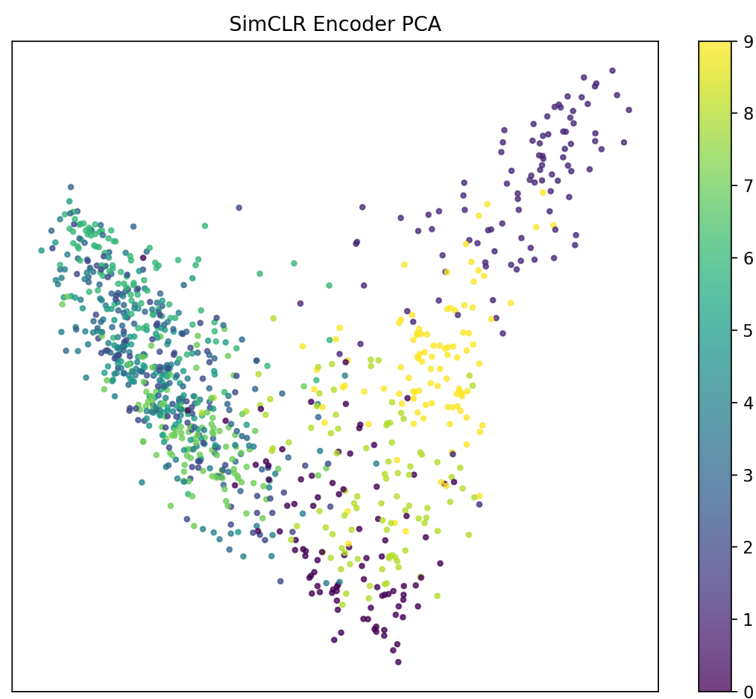


Figure 10: SimCLR Encoder PCA

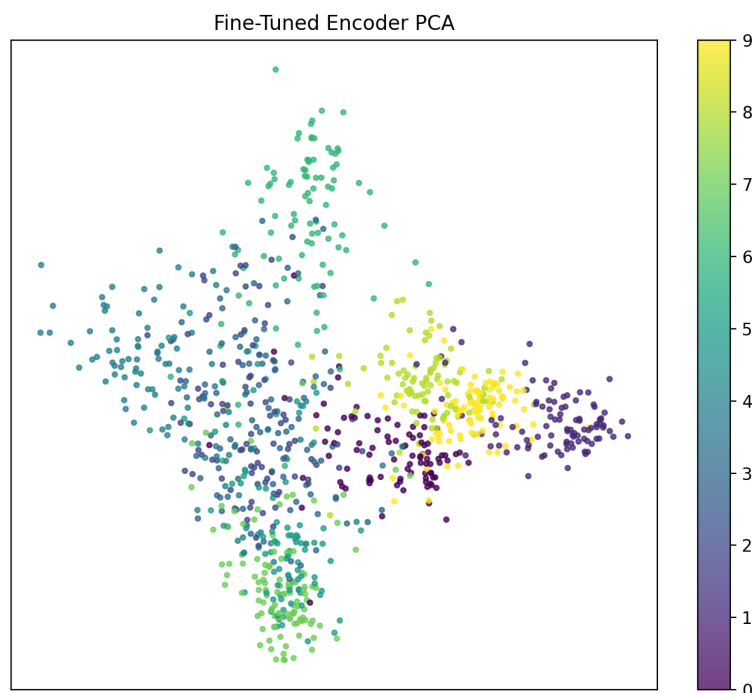


Figure 11: Fine-Tuned Encoder PCA

Do features from the random encoder show class-wise grouping?

No, most classes overlap heavily, and no clear structure is visible

Do features from the SimCLR encoder show better grouping?

Yes, several classes begin forming the distinct regions and clusters

Does fine-tuning improve class separation?

Yes, finetuning produces clearest clusters and best separation among classes

Which classes are still confused?

Some overlap remains between visually similar classes such as dog and cat, horse and deer, truck and automobile, ship and airplane

12 Final Comparison

Method	Test Accuracy
Supervised Baseline (10% Labels)	45.87%
Random Encoder + Linear Probe	26.960%
SimCLR Encoder + Linear Probe	74.660%
SimCLR Fine-Tuning	81.150%

Table 7: Final Performance Comparison

13 Discussion

Results demonstrate effectiveness of self supervised learning, although the SimCLR was trained without labels, it learned highly informative visual representations, so this is evident from similarity analysis, linear probe accuracy, and PCA visualizations. Finetuning further improved the performance by adapting learned features to classification task

14 Implementation Issues and Limitations

- Supervised baseline suffered from overfitting due to limited labeled data
- PCA provides only a 2D projection, and may not fully capture feature relationships
- Certain visually similar classes remained difficult to separate completely
- SimCLR pretraining required substantial GPU computation time

15 AI Usage for learning and guidance

<https://chatgpt.com/share/6a1b883b-a7f0-8320-ae91-631955228361>
<https://chatgpt.com/share/6a1b8879-7638-8323-91f6-688cab9ca053>
<https://chatgpt.com/share/6a1c0229-2dd0-8321-9266-d83242f1b736>
<https://chatgpt.com/share/6a1c0219-b15c-8324-afbb-a6f91a25b78e>